OSNIAS Token — Technical White Paper & Audit Reference

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AUDIT SCOPE PRINCIPLE
OSNIAS is a governance and strategic reserve token. It shares the Sei execution layer with clearing register tokens, but the two economic domains must remain strictly isolated.

Canonical Sei Governance State

OSNIAS governance is intentionally chain-local. Sei EVM is the canonical production blockchain for Osnias Clearing and the sole governance domain for OSNIAS. Only OSNIAS balances held on Sei and eligible under the governance contract rules can contribute voting power.

This design is adopted for simplicity, determinism and auditability. OSNIAS may leave Sei through its omnichain functionality and, on supported EVM chains, may be held in self-custody or cold storage and may be technically integrated by third-party protocols for staking, farming, lending, liquidity provision, collateral, escrow, DEX trading, CEX custody or other compatible DeFi uses. These are interoperability capabilities only and do not constitute protocol promises of yield, liquidity, price appreciation, listing or market support. Cross-chain mobility does not extend governance rights.

> *OSNIAS on Sei = economic value + governance eligibility.*
> *OSNIAS outside Sei = economic value only.*

Governance eligibility is restored only when OSNIAS returns to the canonical Sei chain and is held in a form recognized by the applicable voting mechanism. The protocol does not aggregate governance power from remote chains, external staking systems, farms, custodial representations or other off-Sei environments.

Observable and Deterministic Electorate

Because governance is confined to Sei, the protocol maintains a single observable governance state. At any point in time, the governance system can derive the set of governance-eligible OSNIAS balances and the addresses entitled to exercise or receive delegated voting power from the canonical Sei state, subject to the applicable voting rules and checkpoints. This provides a deterministic voting population without requiring cross-chain reconstruction of off-Sei positions.

This reduces governance complexity and removes the need for cross-chain vote aggregation, remote vote synchronization, bridge-dependent voting messages or reconciliation of duplicated and stale voting power. It also gives auditors a single canonical ledger against which governance eligibility can be tested.

SECURITY / GOVERNANCE PRINCIPLE
Sei is the single source of truth for OSNIAS voting eligibility. An OSNIAS token may circulate economically outside Sei, but it cannot exercise Osnias Clearing governance rights until it returns to the canonical governance chain.

Governance Locality Invariant

> *A token cannot simultaneously provide voting power on Sei and on another blockchain. Economic mobility is omnichain; governance authority is not.*

For audit purposes, this invariant should be tested independently from the OFT transfer mechanism. The reviewer should verify that off-Sei representations cannot produce valid Osnias Clearing voting power and that returning the asset to Sei restores eligibility only through the intended canonical state transition.

OSNIAS is deployed on the same blockchain as the Osnias Clearing register tokens for operational convenience, infrastructure consistency and architectural coherence.

This co-location must never be interpreted as an economic, functional or liquidity relationship between the two asset classes.

OSNIAS and the clearing register tokens are designed as strictly separated systems.

Their coexistence on the same blockchain does not create any right, conversion mechanism, collateral relationship or liquidity bridge between them.

The separation is enforced at the register-token contract level.

Core Separation Invariant

> *OSNIAS must never become directly or indirectly convertible into a clearing register token through the Osnias Clearing protocol.*

Likewise, clearing register tokens must never become directly or indirectly convertible into OSNIAS through protocol-native mechanisms.

The two systems serve fundamentally different purposes:

```
OSNIAS
=
Governance
+ Strategic Reserve
+ Externally Composable EVM Asset
+ Ecosystem Alignment

Clearing Register Tokens
=
Internal Clearing Representation
+ Register Accounting
+ Clearing Settlement Logic
```

They must remain technically and economically independent.

Prohibited Interactions

The protocol architecture explicitly prohibits the following interactions between OSNIAS and any clearing register token.

No Swap

There must be no native swap mechanism between:

```
OSNIAS ⇔ Clearing Register Token
```

Neither the OSNIAS contract nor the clearing register-token contracts should provide a conversion function between the two systems.

No Liquidity Pair

OSNIAS must not be paired with a clearing register token in a protocol-authorized liquidity pool.

The architecture must not intentionally create:

```
OSNIAS / Clearing Token LP
```

or any equivalent liquidity representation.

No Farming Pair

OSNIAS must not be farmed against a clearing register token.

The clearing protocol must not create or authorize yield mechanisms whose economic basis is a combined OSNIAS / clearing-token position.

No DEX Market

The clearing register tokens must not be exposed through protocol-authorized DEX markets against OSNIAS.

The protocol must therefore not establish markets such as:

```
OSNIAS → Clearing Token
Clearing Token → OSNIAS
```

through an AMM, DEX router or protocol-native exchange mechanism.

No CEX Market Supported by the Protocol

Osnias Clearing must not create, sponsor or technically support exchange markets between OSNIAS and the clearing register tokens on centralized exchanges.

No Collateral Dependency

The value, issuance or settlement of a clearing register token must not depend on the market value of OSNIAS.

Similarly, the value or transferable supply of OSNIAS must not depend on the quantity of clearing register tokens in circulation.

Enforcement Boundary

This separation is enforced primarily in the design and code of the clearing register tokens.

The register-token contracts must prevent the clearing units from becoming freely transferable market assets capable of being routed into external liquidity mechanisms.

OSNIAS, by contrast, is intentionally transferable and technically composable with external EVM and DeFi environments on supported networks. Such composability describes possible integrations by third-party protocols and does not constitute a promise by Osnias Clearing of yield, liquidity, listing, market value or price appreciation.

This creates a deliberate asymmetry:

```
OSNIAS:
freely transferable
externally marketable
DEX-compatible
CEX-compatible
farmable
stakeable
lendable
OFT-enabled

Clearing Register Tokens:
restricted by protocol rules
not intended for free market circulation
not swappable against OSNIAS
not usable in OSNIAS liquidity pools
not part of OSNIAS farming strategies
```

External EVM Composability

OSNIAS is designed to remain technically composable outside the canonical Sei governance domain. On supported EVM networks, third-party protocols may choose to integrate OSNIAS for custody, staking, farming, lending, escrow, liquidity provision, collateral arrangements, DEX trading, CEX custody or other compatible applications.

These uses are external technical possibilities. They are not required features of the core token contract and do not create any obligation for Osnias Clearing to provide a market, liquidity venue, yield mechanism, exchange listing or collateral facility.

TECHNICAL COMPOSABILITY ONLY

OSNIAS interoperability and external DeFi compatibility must not be interpreted as a guarantee or representation of yield, liquidity, price support, price appreciation, market depth or availability of any third-party service.

The governance consequence remains deterministic: an OSNIAS balance that is not present on Sei in a form recognized by the applicable voting mechanism has no Osnias Clearing voting power. Returning OSNIAS to the canonical Sei governance chain restores governance eligibility according to the voting rules.

Same Chain Does Not Mean Same Economy

The decision to place OSNIAS and the clearing register tokens on the same production blockchain is purely architectural: Sei provides the common execution environment and the canonical governance state, while the two token classes remain economically isolated.

The shared blockchain provides:

- common execution infrastructure;
- common security environment;
- simpler deployment and monitoring;
- reduced cross-chain operational complexity;
- easier protocol administration;
- coherent technical tooling.

It does not create economic interoperability.

Shared execution layer, separate economic domains.

Auditor Requirement

Auditors should specifically verify that no contract path creates an unintended economic bridge between OSNIAS and the clearing register-token system.

The review should include:

- transfer restrictions of the clearing tokens;
- DEX compatibility restrictions;
- router interaction restrictions;
- approval behavior;
- liquidity-pool exposure;
- farming compatibility;
- collateral hooks;
- internal conversion functions;
- indirect conversion paths;
- privileged administrative functions.

The intended security property is not merely the absence of an explicit swap() function.

The intended property is stronger:

No protocol-controlled or protocol-authorized execution path should allow OSNIAS and a clearing register token to form a direct exchange, liquidity, farming or collateral relationship.

Economic Firewall

This separation can be described as an economic firewall between the governance/reserve layer and the clearing-accounting layer.

```
          OSNIAS ECONOMY
                │
    Governance / Reserve / Markets
                │
                X
          NO CONVERSION
                X
                │
      CLEARING REGISTER ECONOMY
                │
      Accounting / Settlement
```

This firewall is a structural principle of Osnias Clearing and must be preserved in all future protocol upgrades.

Composability Invariant

External EVM use of OSNIAS must not extend, mirror or synthesize Osnias Clearing voting rights outside Sei. External integrations may use the asset economically; governance remains chain-local.

Document status: public pre-audit technical architecture draft for developer and smart-contract review.

Governance-Constrained Management Model

OSNIAS governance is designed to supervise strategic use of the token without replacing day-to-day management. The protocol intentionally avoids an architecture in which every operational decision requires an on-chain vote.

GOVERNANCE PRINCIPLE
Governance should constrain management, not replace management. Smart contracts enforce non-negotiable safety invariants; governance defines strategic mandates and material limits; the Token Manager retains discretion within those approved boundaries.

This model is deliberate. Osnias Clearing cannot predict in advance every future partnership, market condition, infrastructure requirement, liquidity decision or ecosystem opportunity. Excessive hard-coding of management policy would reduce operational adaptability and could make governance itself a source of technical and economic rigidity.

Accordingly, the contract architecture should distinguish immutable safety constraints from discretionary management authority. Governance constrains the latter through reserved matters, mandates, limits and accountability, rather than by attempting to encode every future management decision in advance.

Decision Authority Matrix

Decision Class	Typical Scope	Vote Required	Manager Authority
Routine management	Technical administration, execution, partner coordination, ordinary asset management within approved policy.	No	Full discretion within protocol invariants and approved policies.
Management under mandate	Execution of a previously approved strategic framework, budget, allocation envelope or partnership mandate.	Framework vote only	Manager chooses execution methods, timing and counterparties within the approved mandate.

Decision Class	Typical Scope	Vote Required	Manager Authority
Reserved governance matters	Exceptional use of OSNIAS reserves, major strategic infrastructure investment, fundamental governance changes, or matters expressly reserved to token holders.	Yes	Execution only after a valid governance decision.

Management Flexibility and Code Design

The OSNIAS codebase should remain clear, minimal and auditable. Governance controls should not be scattered across routine functions merely to demonstrate decentralization. Each restriction must correspond to a defined security or governance objective.

The preferred design is therefore to keep a small number of explicit, reviewable invariants and reserved actions, while leaving ordinary management functions operationally flexible where they do not threaten token-holder ownership, supply integrity, cross-chain accounting, governance integrity or the separation from clearing register tokens.

AUDIT EXPECTATION
Auditors should verify that management flexibility does not create hidden authority to mint arbitrary supply, seize third-party balances, bypass governance-reserved matters, break OFT supply conservation, or create any direct or indirect economic bridge between OSNIAS and clearing register tokens.

Non-Negotiable Contractual Invariants

Invariant	Required Property	Management Override
Supply integrity	No arbitrary post-genesis supply creation beyond the contract design.	Not permitted
Holder property rights	Management cannot seize or burn third-party balances merely by administrative decision.	Not permitted
Governance locality	Governance rights are attached to the canonical Sei representation. OSNIAS outside Sei carries economic value but no Osnias Clearing vote.	Not permitted
OFT conservation	Cross-chain movement must preserve the intended global supply accounting.	Not permitted
Clearing-token firewall	No protocol-authorized swap, farming pair, liquidity pool, collateral relationship or conversion path between OSNIAS and a clearing register token.	Not permitted
Reserved matters	Actions explicitly classified as governance-reserved require the applicable vote.	Vote required
Routine management	Operational execution within valid mandates and protocol invariants remains discretionary.	Permitted

Invariant	Required Property	Management Override
Single governance state	Voting eligibility is determined only from the canonical Sei state; no cross-chain vote aggregation or remote voting power is recognized.	Not permitted

Audit Interpretation

The governance architecture should not be evaluated under the assumption that every management action must be voted on. The intended security model is based on separation of powers: immutable contract rules protect core invariants; token-holder governance controls reserved strategic matters; management retains execution authority within those boundaries.

For audit purposes, the critical question is therefore not whether the manager has discretion, but whether that discretion can cross a prohibited boundary without the required governance authorization or violate a hard-coded invariant.

REVIEW QUESTION
Can the Token Manager act flexibly inside the permitted management domain while remaining technically unable to violate the protocol's hard safety boundaries? This is the intended OSNIAS governance design objective.

Security Validation Pipeline

OSNIAS is intentionally simple at the token-contract level: it does not implement native staking, farming, escrow, liquidity pools or clearing-token conversion logic. This reduces the native DeFi attack surface, but it does not remove the need for disciplined security validation.

Slither is used as part of the OSNIAS pre-audit security pipeline to statically analyze contract structure, access-control boundaries, token-supply paths, privileged operations and external-call surfaces. Slither is not treated as an audit, a formal proof, or an on-chain security mechanism.

> *Security principle: static analysis supports verification of the OSNIAS invariants; it does not replace unit tests, invariant testing, cross-chain integration tests or manual security review.*

Security objective	Slither coverage	Validation method	Audit interpretation
Contract exposure and visibility	Strong fit	Slither	Identify public/external functions, inheritance, modifiers and exposed state-changing paths.
Access control / Manager privilege boundaries	Strong fit	Slither + tests	Verify that privileged operations are restricted as intended and that no sensitive path is accidentally public.
Supply integrity / unexpected mint paths	Strong fit	Slither + invariant tests	Map mint/burn paths and verify that no post-genesis mint path exists unless explicitly intended.
Burn permissions	Strong fit	Slither + unit tests	Confirm that burn behavior cannot be used to destroy third-party balances outside the intended authorization model.
External calls / OFT surface	Useful	Slither + integration tests	Review external-call surfaces and OFT-related paths; validate behavior against LayerZero integration tests.
OFT global supply conservation	Partial	Invariant + cross-chain tests	Static analysis can inspect local debit/credit paths, but end-to-end supply conservation must be exercised across chain representations.

Governance locality on Sei	Partial	Architecture + governance tests	Verify that voting power is created and observed only on the canonical Sei governance deployment.
OSNIAS / Clearing Token firewall	Partial	Multi-contract review + tests	Verify that OSNIAS and Clearing Register contracts expose no protocol-controlled conversion, swap, LP, farming or collateral path between the two domains.
External DeFi composability	Out of scope as a guarantee	Third-party protocol review	OSNIAS may be used in external LPs, staking, farming, lending, escrow or collateral systems. Security of those integrations belongs to the external protocols.

OSNIAS-Specific Security Interpretation

OSNIAS itself is not restricted from participating in external liquidity pools or DeFi integrations. Holders and third-party protocols may create OSNIAS liquidity pairs, staking systems, farming strategies, lending markets or other EVM-compatible integrations, subject to the rules and risks of those external systems.

The hard restriction applies only to the boundary between OSNIAS and the Osnias Clearing Register Tokens. The audit objective is therefore not to prevent OSNIAS liquidity. It is to verify that the protocol code does not create or authorize any OSNIAS / Clearing Register Token conversion, swap, liquidity pair, farming relationship, collateral dependency or other economic bridge.

Recommended Pre-Audit Sequence

1. Reproducible Solidity compilation
2. Slither static analysis
3. Unit tests
4. Fuzz and invariant testing
5. Ethereum Sepolia deployment and integration tests
6. LayerZero / OFT cross-chain tests
7. Manual security review against the white-paper invariants
8. Sei mainnet deployment review and configuration verification

> *Slither is a highly recommended component of the OSNIAS pre-audit security pipeline. Protocol-specific invariants are validated through complementary unit, fuzz, integration, cross-chain and manual review procedures.*

Mainnet Privileged-Key Security

The Ethereum Sepolia reference deployment intentionally uses a simplified administrative configuration suitable for development and testing. The production Sei deployment is intended to replace single-key privileged administration with a multisignature security architecture for critical administrative roles.

> *Mainnet security principle: critical administrative authority must not depend on a single externally owned account. Signer separation, cold-key custody, recovery procedures and an external address registry form part of the production operational-security model.*

- Privileged OSNIAS administration: controlled by a production multisignature rather than a single EOA.
- Cold / emergency keys: held separately and used only under defined recovery or emergency procedures.
- Signer resilience: quorum-based control reduces single-key compromise and single-person dependency.
- External registry: authoritative records of production contract addresses, privileged roles and recovery procedures are maintained outside the token contract.
- Test environments: Sepolia and sandbox deployments may retain simplified key management to preserve development speed and testability.

Administrative Domain Separation

The OSNIAS administrative domain remains strictly separated from the Clearing Register Token administrative domain. No OSNIAS privileged role, multisig or manager authority is intended to provide administrative control over any Clearing Register Token.

> *Security invariant: compromise of an OSNIAS privileged role must not, by design, grant administrative authority over Clearing Register Tokens.*

This extends the economic firewall into the operational-security layer: OSNIAS and the Clearing Register Tokens remain separate not only in liquidity, conversion and market behavior, but also in privileged-key authority and administrative control paths.

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